

Pumping Lemma

1. $L = \{0^{2n}1^n \mid n \geq 0\}$ where $\Sigma = \{0, 1\}$

Assuming,

L is regular.

s is a string such that $|s| \geq p$ | $s = 0^{2p}1^p$
 $|s| > p$ and $s \in L$

Decompose,

$$s = xyz$$

$$= 0^a 0^b 0^{p-a-b} 0^p 1^p$$

$$|y| \geq 1$$

$$|xy| \leq p$$

$$s = x(y)^i z$$

for all i s should belong to language.

Let, $i = 2$

$$s = x(y)^2 z$$

$$= 0^a (0^b)^2 0^{p-a-b} 0^p 1^p$$

$$= 0^a 0^{2b} 0^{p-a-b} 0^p 1^p$$

$$= 0^{b+2p} 1^p$$

where, $2p+b > 2p \therefore$ It is not in language

2 $L = \{w \mid w \text{ is not a palindrome}\}$ where $\Sigma = \{0, 1\}$

Assuming, $\bar{L}$ is ~~not~~ a regular language

$\bar{L} = \{w \mid w \text{ is a palindrome}\}$

s is a string such that $|s| \geq p$ | $s = 0^p 1 0^p$
 $s \in \bar{L}$

Decomposing,

$$s = xyz$$

$$|xy| \leq p$$

$$= 0^a 0^b 0^{p-a-b} 1 0^p$$

$$|y| \geq 1$$

$$s = x(y)^i z$$

for all value of i , s should be in language.

let, $i=2$

$$s = x(y)^2 z$$

$$= 0^a (0^b)^2 0^{p-a-b} 1 0^p$$

$$= \cancel{0^a} 0^b \cancel{0^b} 0^p \cancel{0^{p-a-b}} 1 0^p$$

$$= 0^{b+p+p} 1 \notin L$$

As L doesn't belong in language. It's complement

$\bar{L} = L$ will also not belong in language.

3. $L = \{a^n \text{ where } n \text{ is prime}\}$, $\Sigma = \{a\}$

Assuming,

L is regular

s is a string such that $|s| \geq p$, $s \in L$

$s = a^q$, q must be a prime number

$$|s| = q \geq p$$

Decomposing,

$$|xy| \leq p, |y| \geq 1$$

$$s = xyz$$

$$= x(y)^i z$$

$$= a^m (a^k)^i a^{p-m-k} a^{q-p}$$

[for all i s should belong in language]

$$= \cancel{a^m} a$$

$$\text{let } i = q+1$$

$$* S = a^m a^{kq+k} a^{p-m-k} a^{q-p}$$

$$= \cancel{a^m} a^{kq} \cancel{a^k} \cancel{a^p} \cancel{a^{-m}} \cancel{a^{-k}} a^q \cancel{a^{-p}}$$

$$= a^{kq+q} = a^{q(k+1)}$$

$$\therefore |s| = q(k+1) \text{ where } q \geq 2 \text{ and } k \geq 1$$

$$\therefore (k+1) \geq 2$$

$\therefore q(k+1) \rightarrow$ product of two integers and it cannot be a prime number

$$\text{as } q(k+1) \neq \text{number} \therefore s \notin L$$

$\therefore L$ is not a regular language

$$\underline{4} \quad L = \{a^m b^n \text{ where } m \neq n\}, \Sigma = \{a, b\}$$

Assuming, L is regular language

$\$S$ is a string such that $|s| \geq p$ and $s \in L$

$$\therefore s = a^p b^{p+p!}$$

Decomposing,

$$s = xyz \quad |xy| \leq p \text{ and } |y| \geq 1$$

$$= \cancel{a^p} a^q a^r a^{p-q-r} \cancel{a} b^{p+p!}$$

$$s = x(y)^i z$$

$$= a^q (a^r)^i a^{p-q-r} b^{p+p!}$$

$$= a^{p+ir-r} b^{p+p!}$$

xy^iz will be in language if

$$p + ir - r \neq p + p!$$

$$\Rightarrow ir - r \neq p!$$

$$\Rightarrow i = \frac{p!}{r} + 1$$

if $i = \frac{p!}{r} + 1$, it will not be in language.

$$\text{Let } i = \frac{p!}{r} + 1$$

$$s = xy^iz$$

$$= a^q (a^r)^{\frac{p!}{r} + 1} a^{p-q-r} b^{p+p!}$$

$$= a^q a^{p!+r} a^{p-q-r} b^{p+p!}$$

$$= \cancel{a^q} \cancel{a^{p!}} \cancel{a^r} \cancel{a^p} \cancel{a^q} \cancel{a^r} b^{p+p!}$$

$$= a^{p+p!} b^{p+p!}$$

$$\text{As, } a^{p+p!} b^{p+p!} \notin L$$

$\therefore L$ is not regular language.

#CFG

~~1~~ 1 $L = \{0^n 1^n, \text{where } n \text{ is odd}\}, \Sigma = \{0, 1\}$
 $S \rightarrow 00S11 \mid 01$

2 $L = \{\text{the length of } w \text{ is divisible by } 3\}, \Sigma = \{0, 1\}$

$S \rightarrow ABCS \mid \epsilon$

$A \rightarrow 011$

$B \rightarrow 011$

$C \rightarrow 011$

3 $L = \{w_1 \# w_2, \text{the number of } 0\text{'s in } w_1 \text{ is equal to number of } 1\text{'s in } w_2\}$

$S \rightarrow 0S1 \mid 1S \mid S0 \mid \#$

4 $L = \{w = 0^i 2^j 1^k, \text{where } i=k, i, k \geq 1 \text{ and } j \geq 2\}$
 $\Sigma = \{0, 1, 2\}$

$w = 0^i 2^j 1^k$

$S \rightarrow 0S1 \mid 0A1$

$A \rightarrow 2A \mid 22$

5 $L = \{w = 0^i 2^j 1^k, \text{where } i \text{ is multiple of } 2, k \text{ is two more than multiple of } 3, j = k+1 \text{ and } i, j, k \geq 0\}$

$w = 0^i 2^{i+k} 1^k$
 $= 0^i 2^i 2^k 1^k$

$S \rightarrow AB$

$A \rightarrow 00A22 \mid \epsilon$

$B \rightarrow 222B111 \mid 2211$

6 | $L = \{ \text{the no. of 0s and 1s are diff in } w \}$

$$S \rightarrow AB$$

$$A \rightarrow T0A \mid T0T$$

$$B \rightarrow T1B \mid T1T$$

$$T \rightarrow 0T1T \mid 1T0T \mid \epsilon$$

Regular expression to CFG

$$(a^*b^* + (ac + b^*c)b)^*$$

CFG,

$$S \rightarrow TS \mid \epsilon$$

$$T \rightarrow A \mid B$$

$$A \rightarrow PQ$$

$$P \rightarrow aP \mid \epsilon$$

$$Q \rightarrow bQ \mid \epsilon$$

$$B \Rightarrow Cb$$

$$C \rightarrow ac \mid Qc$$

Derivation, Parse Tree and Ambiguity

a) 01101010

$$S \rightarrow 0S0$$

$$\rightarrow 01S10$$

$$\rightarrow 01A10$$

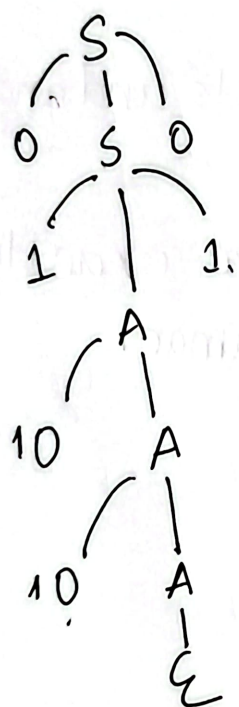
$$\rightarrow 0110A10$$

$$\rightarrow 011010A10$$

$$\rightarrow 011010E10$$

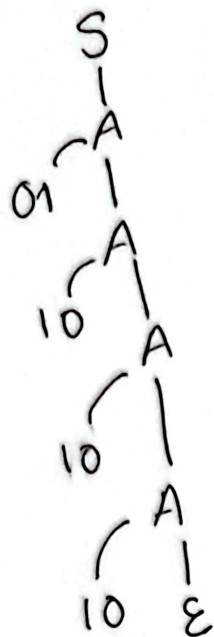
$$\rightarrow 01101010$$

b)

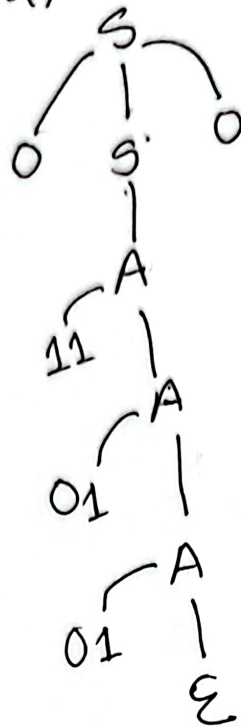


c) String = 011010

PT1,



PT2,



∴ the given tree grammar is ambiguous

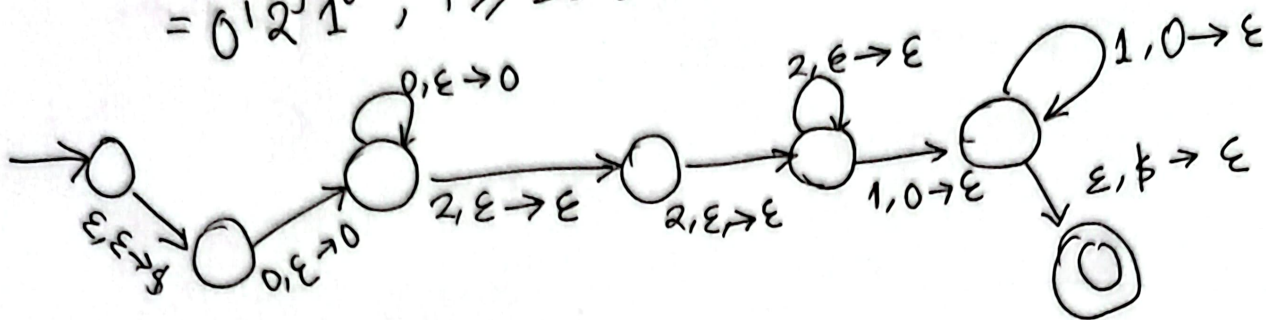
d) All four letter strings that starts and ends with different symbol.

∴ 8 four letter strings will have exactly one parse tree in the given grammar.

PDA

$$1. w = 0^i 2^j 1^k$$

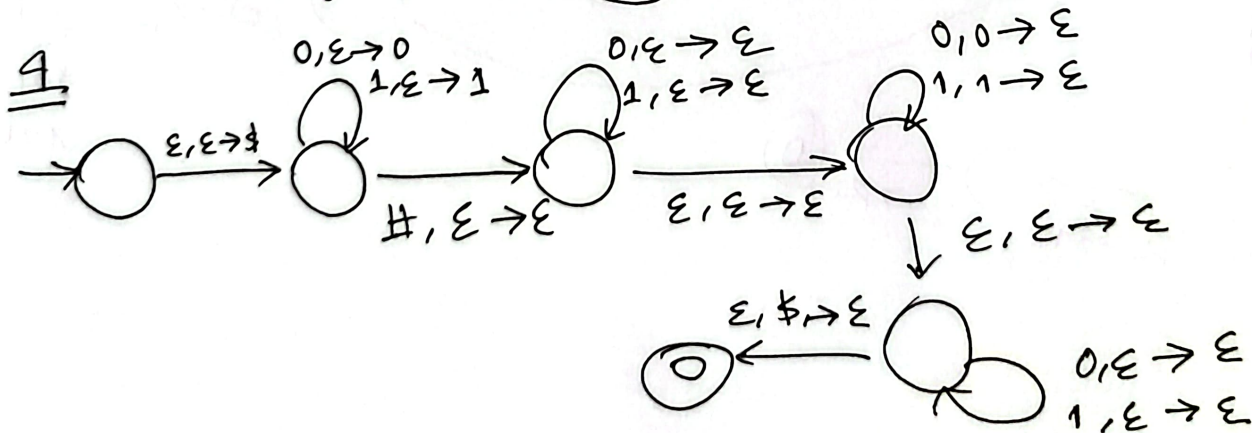
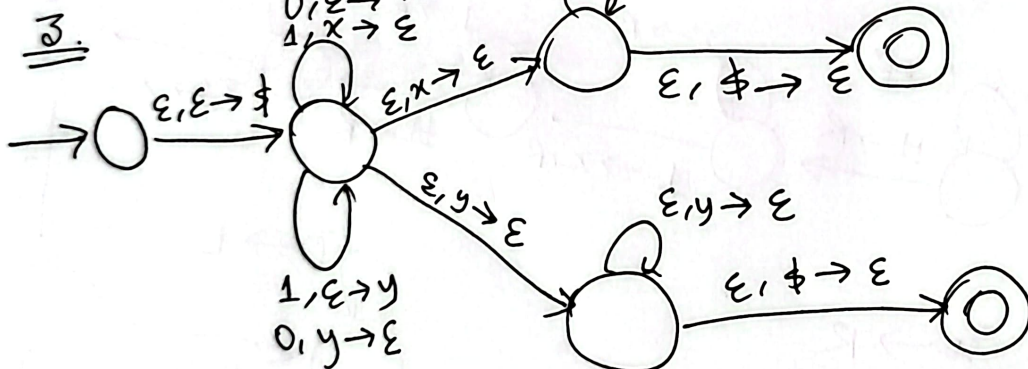
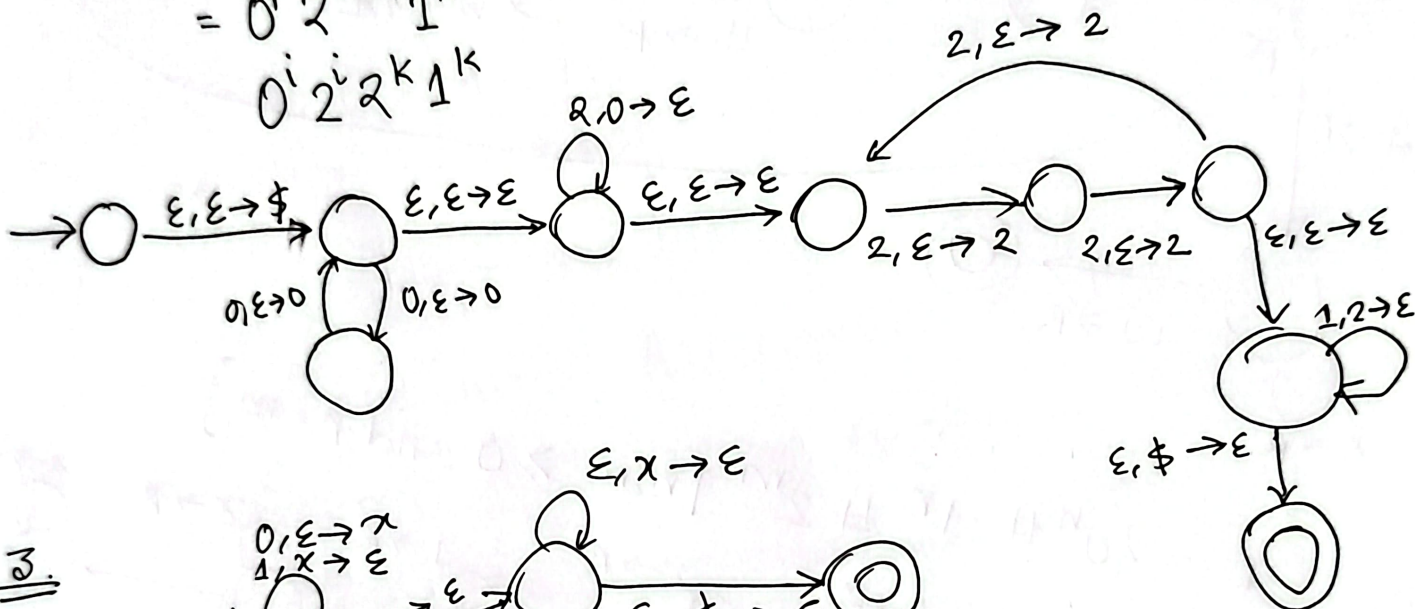
$$= 0^i 2^j 1^i, i \geq 1, j \geq 2$$



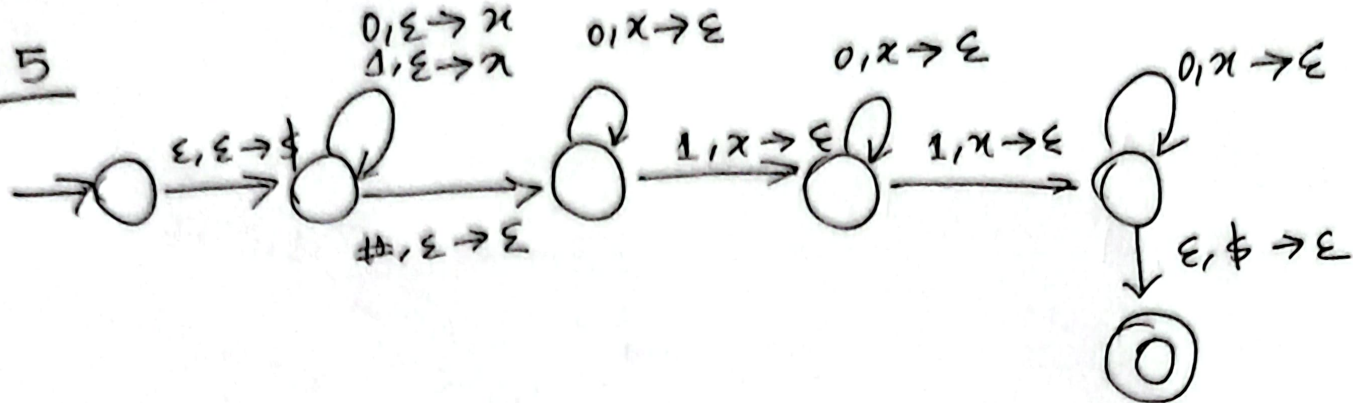
2. $w = 0^i 2^j 1^k$

$$= 0^i 2^{i+k} 1^k$$

$$0^i 2^i 2^k 1^k$$

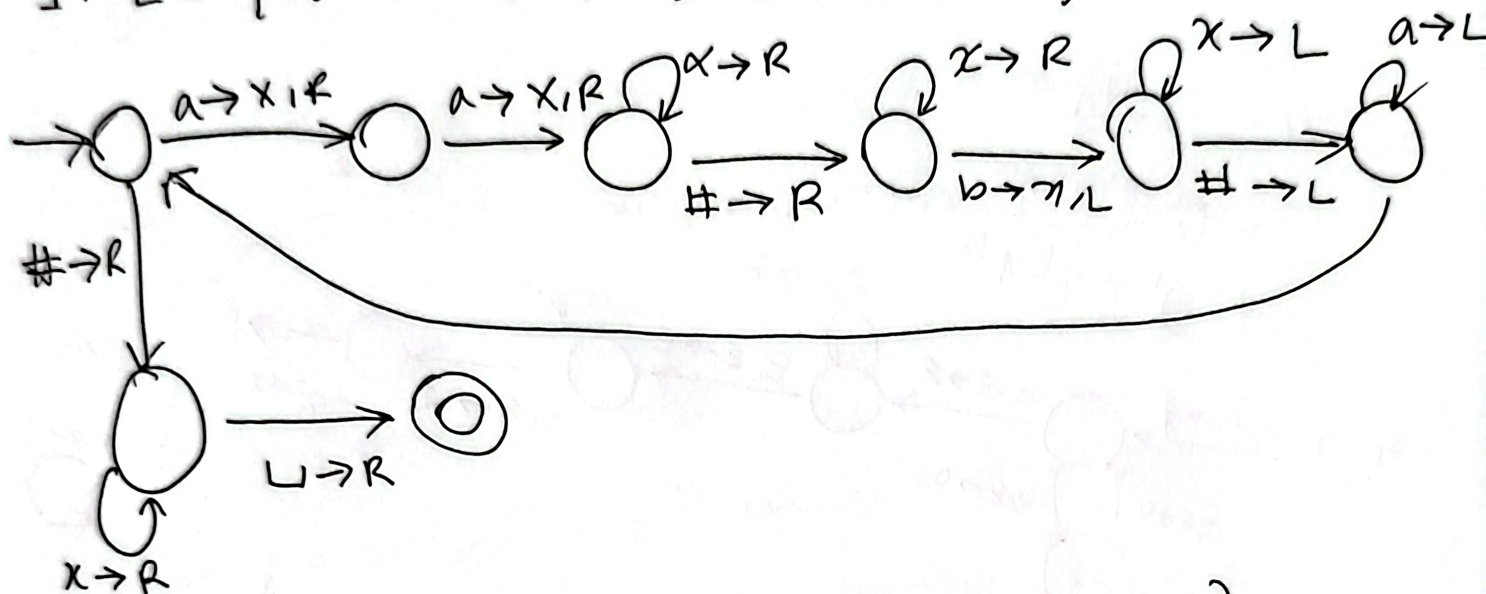


5

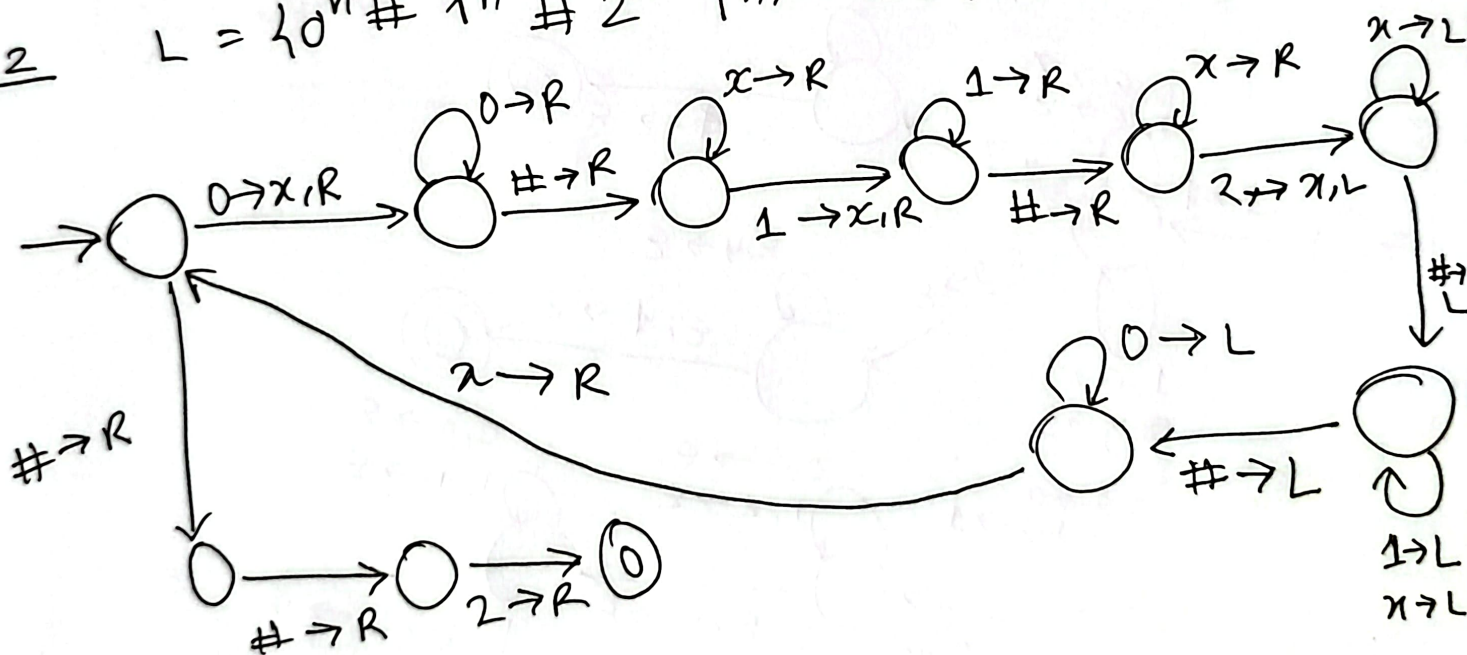


Turing machine

1. $L = \{a^{2n} \# b^n \mid n \geq 0\}$, $\Sigma = \{a, b, \#\}$



2. $L = \{0^n \# 1^n \# 2^m \mid n, m \geq 0 \text{ and } n < m\}$



3 $A_{DFA} = \{ \langle D, w \rangle \mid D \text{ is a DFA that accepts input string } w \}$

Let, M be a Turing machine that decides

A_{DFA} on input $\langle D, w \rangle$ where $D = (Q, \Sigma, \delta, q_0, F)$

The machine works as follows,

i) start at the initial state q_0 of the DFA D .

ii) Read w by symbol from $L \rightarrow R$

iii) for each state in w , compute next state by transition function,

$$q \leftarrow \delta(q, a)$$

iv) Continue until all symbols of w are processed

v) If curr state belong to the set of accepting states F , then accept otherwise Reject.